

Multiple Choice (10 marks)

1. The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?
 - (a) Magnetic susceptibility
 - (b) Hall coefficient
 - (c) Electrical resistivity
 - (d) Thermal conductivity
2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
 - (a) 2
 - (b) 250
 - (c) 5,000
 - (d) 10,000
3. Which of the following ions CANNOT be used as a dopant in germanium to make an n-type semiconductor?
 - (a) As
 - (b) P
 - (c) Sb
 - (d) B
4. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T . The average total energy of the oscillator is
 - (a) $(1/2) k T$
 - (b) kT
 - (c) $(3/2) k T$
 - (d) $3 kT$
5. An organ pipe, closed at one end and open at the other, is designed to have a fundamental frequency of C (131 Hz). What is the frequency of the next higher harmonic for this pipe?
 - (a) 44 Hz
 - (b) 196 Hz
 - (c) 262 Hz
 - (d) 393 Hz

True / False (10 marks)

Answer True or False

6. True or False: A particle moving at $v=0.60\ c$ will travel only 288 m before decaying, based on its rest frame decay time.
7. True or False: The eigenvalues of a Hermitian operator can be either real or imaginary numbers.
8. True or False: The resolving power of a grating spectrometer that can resolve wavelengths of 500 nm and 502 nm is 250.
9. True or False: The energy required to ionize helium by removing one electron is 24.6 eV.
10. True or False: The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately $1.0\ \text{GeV}/c^2$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given a set of 10 one-second measurements of a radioactive isotope's disintegration, analyze how long the student should measure to achieve a 1% uncertainty in the rate.
12. Discuss the advantages of using dye lasers for spectroscopy across visible wavelengths, including their tunability and applications in various scientific fields.

End of Paper